

4.12.25

не было

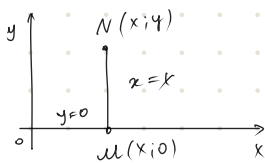
Сформи конспект.

Полный дифференциал. Независимость криволинейн. инт. от пути интегр.

$$\textcircled{1} \quad \int_{A(1,1)}^{B(2,3)} (x+3y)dx + (y+3x)dy = \int_1^2 (x+3)dx + \int_1^3 (y+6)dy =$$

$$= \left(\frac{x^2}{2} + 3x \right) \Big|_1^2 + \left(\frac{y^2}{2} + 6y \right) \Big|_1^3 = 20,5$$

$$\textcircled{2} \quad \text{Найти } U, \text{ если } dU = \underbrace{[y + \ln(x+1)]}_{P(x,y)} dx + \underbrace{(x+1-e^y)}_{Q(x,y)} dy$$



$$\frac{\partial P}{\partial y} = 1 = \frac{\partial Q}{\partial x}$$

$$dx = d(x+1)$$

$$U(x,y) = \int_0^x 0 + \ln(x+1) dx + \int_0^y (x+1-e^y) dy = \left\{ \begin{array}{l} \int \frac{\ln x}{u} \frac{dx}{dv} = x \ln x - \sqrt{x} \cdot \frac{1}{x} dx = \\ du = \frac{1}{x} \quad v = x \end{array} \right.$$

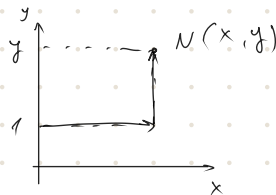
$$= x \ln x - x + C \left\{ = [(x+1) \ln(x+1)] \Big|_0^x + (xy + y - e^y) \Big|_0^y = (x+1) \ln(x+1) - (x+1) + \right.$$

$$+ xy + y - e^y + C$$

$$\textcircled{3} \quad U - ?$$

$$dU = \left(\frac{1}{x} + \frac{1}{y} \right) dx + \left(\frac{2}{y} - \frac{x}{y^2} \right) dy$$

$P(x,y) \qquad Q(x,y)$



$$\frac{\partial P}{\partial y} = -\frac{1}{y^2} \qquad \frac{\partial Q}{\partial x} = -\frac{1}{y^2}$$

$$\int_1^x \left(\frac{1}{x} + \frac{1}{y} \right) dx + \int_1^y \left(\frac{2}{y} - \frac{x}{y^2} \right) dy = \left(\ln x + \frac{x}{y} \right) \Big|_1^x + \left(2 \ln y + \frac{x}{y} \right) \Big|_1^y =$$

$$= \ln \frac{x}{y} - 0 - \frac{1}{y} + 2 \ln y + \frac{x}{y} - 0 - x + c = \ln x + \frac{x}{y} + \frac{x-1}{y} + \frac{x-1}{y} + 2 \ln y + c$$

④ Берем

$$\int_K x dy + y dx = \int_0^{2\pi} (\cos t + \cos t - \sin t + \sin t) dt =$$

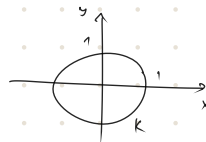
$$y = \cos t \quad dy = -\sin t dt$$

$$x = -\sin t \quad dx = -\cos t dt$$

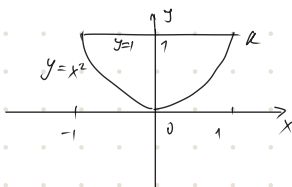
$$0 \leq t \leq 2\pi.$$

1. По параметризации: $x = \cos t$ $z = 1$
 $y = \sin t$

$$= \int_0^{2\pi} \cos 2t dt = \frac{\sin 2t}{2} \Big|_0^{2\pi} = 0$$



2. По $y = x^2$
 $y = 1$



$$\int_K x dy + y dx = \int_{-1}^1 (x \cdot 2x + x^2) dx + \int_1^{-1} 1 dx = \frac{2x^3}{3} \Big|_{-1}^1 + x \Big|_1^{-1} = 1 - (-1) - 1 - (-1) = 0$$

$$y = x^2$$

$$dy = 2x dx$$

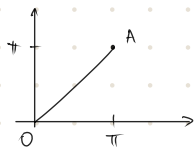
⑤ $\int_{(0,0)}^{(\pi,\pi)} (x+y) dx + (x-y) dy$

1) по прямой OA

$$\int_0^{\pi} (x+x) + (x-x) dx = \int_0^{\pi} 2x dx = 2 \frac{x^2}{2} \Big|_0^{\pi} = \pi^2$$

$$y = x$$

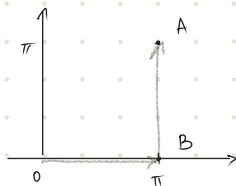
$$dy = dx$$



2) по кривой $y = x + \sin x \Rightarrow dy = 1 + \cos x$

$$\int_0^{\pi} (x + \sin x) dx + \int_0^{\pi} (x - \sin x) (1 + \cos x) dx = \int_0^{\pi} (2x - \sin x \cos x) dx = \int_0^{\pi} (2x - \frac{1}{2} \sin 2x) dx = \left(\frac{x^2}{2} - \frac{1}{2} \cos 2x \cdot \frac{1}{2} \right) \Big|_0^{\pi} = \pi^2 - \frac{1}{4} (1 - 1) = \pi^2$$

3) по прямой OBA ~~BA~~ B($\pi; 0$)

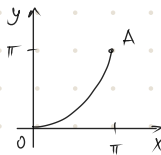


$$\int_0^{\pi} (x+0) dx + \int_0^{\pi} (x-0) dy + \int_0^{\pi} (\pi+y) dx + \int_0^{\pi} (\pi-y) dy = \frac{x^2}{2} \Big|_0^{\pi} + (\pi y - \frac{y^2}{2}) \Big|_0^{\pi} = \frac{\pi^2}{2} + (\pi^2 - \frac{\pi^2}{2}) = \pi^2$$

4) по парабол $y = \frac{x^2}{\pi}$

$$y = \frac{x^2}{\pi}$$

$$dy = \frac{2x}{\pi} dx$$



$$\int_0^{\pi} (x + \frac{x^2}{\pi}) dx + \int_0^{\pi} (x - \frac{x^2}{\pi}) (\frac{2x}{\pi}) dx = \int_0^{\pi} (x + \frac{x^2}{\pi} + \frac{2x^2}{\pi} - \frac{2x^3}{\pi^2}) dx = \int_0^{\pi} (x + \frac{3x^2}{\pi} - \frac{2x^3}{\pi^2}) dx = \left(\frac{x^2}{2} + \frac{3x^3}{\pi \cdot 3} - \frac{2x^4}{\pi^2 \cdot 4} \right) \Big|_0^{\pi} = \frac{\pi^2}{2} + \frac{\pi^3}{\pi} - \frac{\pi^4}{\pi^2 \cdot 2} = \frac{2\pi^2}{2} = \pi^2$$

Решения Тренин:

$$\oint_C P(x,y) dx + Q(x,y) dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy =$$

$$= 4 \int_0^{\pi/2} \int_0^a (y' + x') dx dy = \left| \begin{array}{l} x = r \cos \varphi \\ y = r \sin \varphi \end{array} \right|_{\substack{0 \leq r \leq a \\ 0 \leq \varphi \leq \pi/2}} =$$

$$= 4 \int_0^{\pi/2} \int_0^a (r^2 \cos^2 \varphi + r^2 \sin^2 \varphi) r dr d\varphi = 4 \int_0^{\pi/2} \int_0^a r^3 dr d\varphi = \int_0^{\pi/2} r^4 \Big|_0^a d\varphi = a^4 \int_0^{\pi/2} d\varphi =$$

$$= a^4 \cdot \varphi \Big|_0^{\pi/2} = \frac{a^4 \pi}{2}$$

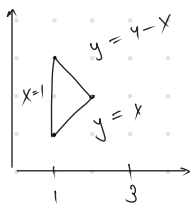
$$\textcircled{6} \oint \underbrace{2(x^2 + y^2)}_P dx + (x+y)^2 dy$$

$$\frac{\partial P}{\partial y} = 4y$$

$$\frac{\partial Q}{\partial x} = 2(x+y)$$

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 2(x+y) - 4y = 2(x-y)$$

C-контур $\triangle LMN$: $L(1; 1)$; $M(2; 2)$, $N(1; 3)$



$$= \iint_D 2(x-y) dx dy = 2 \int_1^2 dx \int_x^{4-x} (x-y) dy =$$

$$= 2 \int_1^2 dx \left(xy - \frac{y^2}{2} \right) \Big|_x^{4-x} = 2 \int_1^2 dx \left(x(4-x) - \frac{(4-x)^2}{2} - \left(\frac{4-x}{2} \right)^2 - \frac{x^2 + \frac{x^2}{2}}{2} \right) dx =$$

$$= 2 \left(\frac{4x^2}{2} - \frac{3}{2} \frac{x^3}{3} - \frac{(x-4)^3}{3 \cdot 2} \right) \Big|_1^2 = 2 \left(6 - 4 + \frac{1}{2} + \frac{4}{3} - \frac{9}{2} \dots \right) = 4 + \frac{4}{3}$$

$$\textcircled{7} \oint_C -\frac{P}{x^2 y} dx + xy^2 dy = \iint_D (x^2 + y^2) dx dy \ominus$$

↑ контур $x^2 + y^2 = R^2$ против часовой стрелки.

$$\frac{\partial Q}{\partial x} = y^2$$

$$\frac{\partial P}{\partial y} = -x^2$$

$$\begin{cases} x = \rho \cos \varphi \\ y = \rho \sin \varphi \\ x^2 + y^2 = \rho^2 \end{cases}$$

$$\ominus \iint_D \rho^2 \cos^2 \varphi + \rho^2 \sin^2 \varphi d\rho d\varphi = \iint_D \rho^2 d\rho d\varphi = \int_0^{2\pi} d\varphi \int_0^R \rho^2 d\rho =$$

$$\begin{matrix} 0 \leq \rho \leq R \\ 0 \leq \varphi \leq 2\pi \end{matrix}$$

$$= \int_0^{2\pi} d\varphi \frac{\rho^4}{4} \Big|_0^R = \frac{R^4}{4} \varphi \Big|_0^{2\pi} = \frac{2\pi R^4}{4} = \frac{\pi R^2}{2}$$

$$\textcircled{8} S = \frac{16}{2\sqrt{c}} x dy - y dx \ominus$$

Сампоугол - ?



$$\begin{cases} x = a \cos^3 t \\ y = a \sin^3 t \end{cases}$$

$$x^{2/3} + y^{2/3} = a^{2/3}$$

$$\begin{aligned} \sin^2 t \cos^2 t &= \left(\frac{\sin 2t}{2} \right)^2 = \\ &= \frac{1}{4} - \frac{1 - \cos 4t}{2} \end{aligned}$$

$$\begin{aligned} dy &= a \cdot 3 \sin^2 t \cos t \, dt \\ dx &= a \cdot 3 \cos^2 t \cdot (-\sin t) \, dt \end{aligned}$$

$$\Leftrightarrow \frac{1}{2} \int (a \cos^3 t \cdot a \cdot 3 \sin^2 t \cdot \cos t - a \sin^3 t \cdot a \cdot 3 \cos^2 t \cdot (-\sin t)) \, dt =$$

$$= \frac{3a^2}{2} \int_0^{2\pi} (\cos^4 t \sin^2 t + \sin^4 t \cos^2 t) \, dt = \frac{3a^2}{2} \int_0^{2\pi} \sin^2 t \cos^2 t \, dt =$$

$$\frac{\sin^2 2t}{4}$$

$$= \frac{3a^2}{16} \int_0^{2\pi} (1 - \cos 4t) \, dt = \frac{3a^2}{16} \cdot 2\pi = \frac{3a^2 \pi}{8}$$

